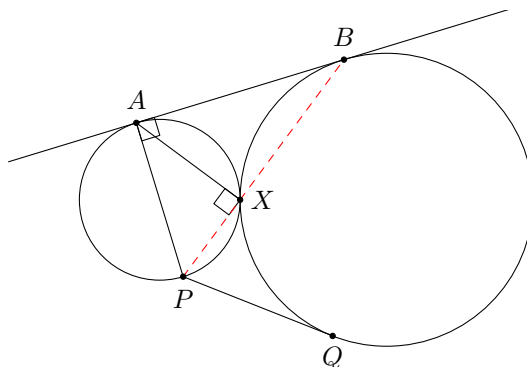


Problem 1 (Lithuania TST 2015). Two circles ω_1 and ω_2 are externally tangent to each other. A common external tangent of these two circles touches ω_1 at A and ω_2 at B . Segment AP is the diameter of the circle ω_1 , and the line passing through point P is tangent to ω_2 at point Q .

Prove that $AP = PQ$.

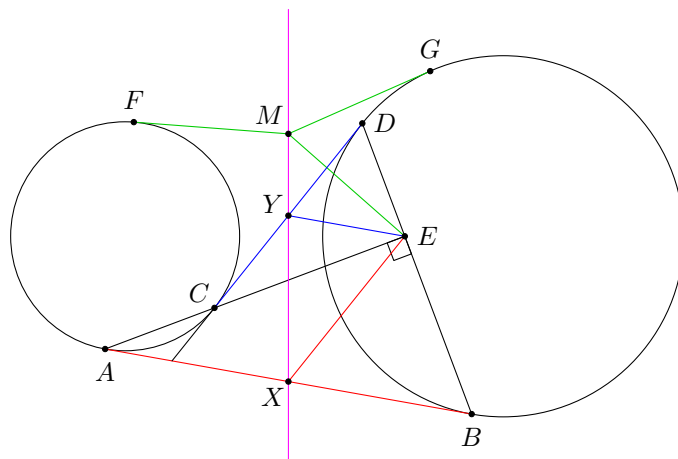


Solution. Let X be the touching point of the two circles. Then since P is the image of B under the negative homothety with centre X sending ω_2 to ω_1 , P, X, B are collinear. Now by power of a point, we have $PQ^2 = PX \cdot PB$. On the other hand, since AP is a diameter we have $\angle AXP = 90^\circ$ and $\angle PAB = 90^\circ$, giving $\triangle PAX \sim \triangle PBA$. So $\frac{PX}{PA} = \frac{PA}{PB}$, which implies $PA^2 = PX \cdot PB$. Therefore $PA = PQ$ as desired. $\square$

Problem 2 (CGMO 2015). Let Γ_1 and Γ_2 be two non-overlapping circles. Suppose

- Points A, C are on Γ_1 and points B, D are on Γ_2 such that AB is a common external tangent to the two circles, and CD is a common internal tangent to the two circles,
- AC and BD meet at E ,
- F is a point on Γ_1 ,
- the tangent line to Γ_1 at F meets the perpendicular bisector of EF at M , and
- MG is a line tangent to Γ_2 at G .

Prove that $MF = MG$.

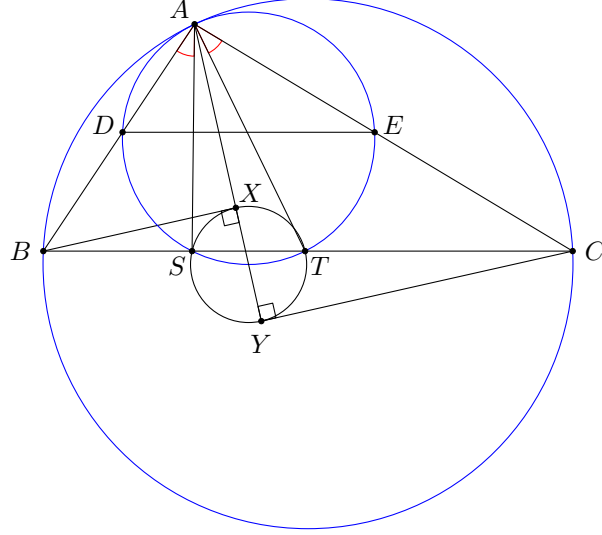


Solution. We only consider the case when C is closer to the line AB than D , since the other case is similar. Denote by X and Y the midpoints of AB and CD respectively. Note that $\angle AEB = \angle ECD + \angle CDE = \angle BAE + \angle EBA = \angle DEC$. So $AC \perp BD$. Now we have $XA^2 = XB^2 = XE^2$ and $YC^2 = YD^2 = YE^2$, so X and Y have the same power to Γ_1, Γ_2 and point circle E . Therefore line XY is the common radical axis of Γ_1, Γ_2 and point circle E .

Now MF is tangent to Γ_1 and $MF = ME$, so the power from M to Γ_1 and point circle E is $MF^2 = ME^2$, i.e. M lies on the radical axis of Γ_1 and point circle E . So M lies on XY , which is the radical axis of Γ_1 and Γ_2 too. Hence $MF^2 = MG^2$, which gives $MF = MG$ as desired. $\square$

Problem 3 (ELMO 2016 modified). In $\triangle ABC$ with $AB \neq AC$, points X and Y are projections of B and C , respectively, to the internal angle bisector of $\angle A$. Let S and T be distinct points on side BC such that $\angle XSY = \angle XTY = 90^\circ$.

Show that the circumcircles of $\triangle AST$ and $\triangle ABC$ are tangent to each other.

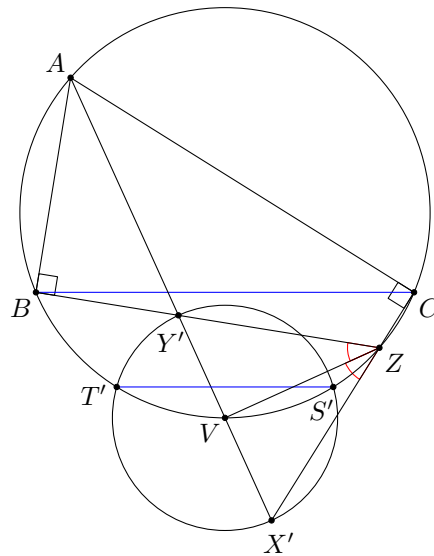


Solution 1. We note that BX and CY are tangent to the circle $(XYST)$, so by length chasing we have

$$\left(\frac{AB \sin(A/2)}{AC \sin(A/2)} \right)^2 = \left(\frac{BX}{CY} \right)^2 = \frac{BS \cdot BT}{CS \cdot CT} = \left(\frac{AB \cdot \sin \angle BAS}{AC \cdot \sin \angle SAC} \right) \left(\frac{AB \cdot \sin \angle BAT}{AC \cdot \sin \angle TAC} \right),$$

which clearing everything gives $\left(\frac{\sin \angle BAS}{\sin \angle SAC} \right) \left(\frac{\sin \angle BAT}{\sin \angle TAC} \right) = 1$. Therefore AS and AT are isogonal, that is, $\angle BAS = \angle TAC$. Now let AB and AC meet the circumcircle of $\triangle AST$ again at D and E . Then from $\angle DAS = \angle TAE$ we have $DE \parallel ST$. Hence A is the positive homothety centre sending D to B and E to C , so it also sends the circle $(ADSTE)$ to the circle (ABC) . As a result, these two circles are tangent at A . $\square$

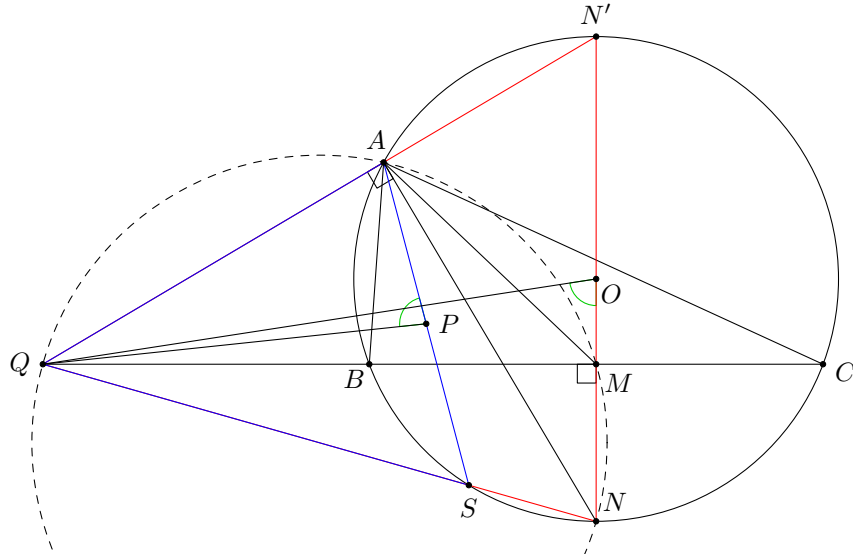
Solution 2. Here we present a solution using the $\sqrt{bc}$ -inversion.



Consider the $\sqrt{bc}$ -inversion, that is, invert about a circle with centre A and radius $\sqrt{AB \cdot AC}$ first, then reflect about the A -angle bisector. Then X' and Y' are on the A -angle bisector such that $\angle ACX' = \angle AXB = 90^\circ$ and $\angle ABY' = \angle AYC = 90^\circ$. By symmetry, the circle $(XYST)$ must be mapped to a circle with diameter $X'Y'$, and S', T' would be the intersection of this circle with the circumcircle of $\triangle ABC$. To prove the assertion is to prove that $S'T' \parallel BC$.

Let Z be the intersection of BY' and CX' , so the right angles imply Z lying on the circumcircle of $\triangle ABC$. Also let V be the midpoint of $\widehat{BC}$ not containing A . Then ZV would be the external angle bisector of $\angle BZC$, i.e. ZV bisects $\angle Y'ZX'$. By $\angle ZX'Y' = 90^\circ - \angle X'AC = 90^\circ - \angle BAY' = \angle ZY'X'$, $\triangle ZY'X'$ is isosceles with $ZY' = ZX'$. Therefore for ZV to bisect $\angle Y'ZX'$, V must be the midpoint of $X'Y'$. Hence $(X'Y'S'T')$ is a circle with centre V . Since V is the midpoint of $\widehat{BC}$, so $S'T' \parallel BC$ by symmetry, as desired. $\square$

Problem 4 (ELMO 2012 Shortlist). Let $\triangle ABC$ be an acute triangle with $AB < AC$ and circumcentre O . Let Q be the intersection of the external bisector of $\angle A$ with BC , and let P be a point in the interior of $\triangle ABC$ such that $\triangle BPA$ is similar to $\triangle APC$. Show that $\angle QPA + \angle OQB = 90^\circ$.



Solution. Since P is the dumpty point, P is the midpoint of AS , where S lies on the circumcircle of $\triangle ABC$ such that AS is the A -symmedian. Denote by M , N and N' the midpoints of BC , $\widehat{BC}$ that does not contain A and $\widehat{BC}$ that contains A . Note that Q, A, N' are collinear. As AN is the internal angle bisector of $\angle BAC$, we have $\angle QAN = \angle QMN = 90^\circ$, so A, Q, N, M are concyclic.

Now we notice that Q, S, N are collinear by angle chasing: since $\angle QNM = 180^\circ - \angle QAM$ and $\angle SNN' = 180^\circ - \angle SAN'$, together with the fact that AS and AM are isogonal in $\angle BAC$ giving $\angle QAM = \angle SAN'$, we have $\angle QNM = \angle SNM$ as desired.

We now have Q as the intersection of AN' and SN , so we realise $\triangle AQS \sim \triangle NQN'$. As P and O are the midpoints of AS and NN' respectively, we have $\triangle QAP \sim \triangle QNO$, implying $\angle QPA = \angle QON = 90^\circ - \angle OQB$ and we are done. $\square$